

Tests: Questions

Test 01: Differential Expression Analysis

Goal: This test focuses on the main differential expression analysis workflow in the project.

Students should practice:

- reading DEG result files
- creating new columns from existing columns
- using logical conditions to classify genes
- reproducing the main ideas from the DEG and volcano scripts

Question 1

Inspect the DEG result table.

Tasks:

- print `dim(deg_res)`
- print the first 6 column names
- show the first 3 rows of `deg_res[, 1:5]`

Hint: You can use functions such as `dim()`, `names()`, and `head()`. Column subsetting with `[rows, columns]` may also be useful.

```
# TODO
```

Question 2

Create a new column called `log10FDR`.

Rule:

- `log10FDR = -log10(adj.P.Val)`

Hint: You can create a new column with the column access operator, use `log10()` for the transformation, and use logical conditions with `&` to classify rows. `table()` can help summarize category counts.

```
# TODO
```

Question 3

Keep only significant genes and show the top 5 genes with the largest positive logFC.

Tasks:

- create `deg_only` using `DEG != "NotSig"`
- print `dim(deg_only)`
- print the top 5 rows sorted by `desc(logFC)`

Hint: Functions from `dplyr`, such as `filter()` and `arrange()`, are useful here. You can use `desc()` for decreasing order and `head()` to keep only a few rows.

```
# TODO
```

Short Answer

In 2 to 3 sentences, explain:

- what the `DEG` column means
- why we classify genes into `Up`, `Down`, and `NotSig` before plotting

Test 02: Enrichment Analysis

Goal: This test focuses on the GSEA workflow in the project.

Students should practice:

- reading GSEA result tables
- extracting important columns
- comparing different GSEA outputs
- explaining what a GSEA result means in simple language

Question 1

Inspect the three GSEA result tables.

Tasks:

- print `dim(gsea_go)`
- print `dim(gsea_kegg)`
- print `dim(hallmark)`
- print the first 3 descriptions from `gsea_go`

Hint: Use `dim()` to inspect table size. Use the column access operator to access one column, and `head()` to show the first few values.

```
# TODO
```

Question 2

Extract key columns from GSEA GO and GSEA KEGG results.

Tasks:

- from `gsea_go`, keep `Description`, `NES`, `p.adjust`, and `setSize`
- from `gsea_kegg`, sort by `p.adjust` from small to large
- print the first 5 pathway descriptions from the sorted GSEA KEGG table

Hint: `select()` can keep specific columns, and `arrange()` can sort rows. Access the `Description` column when you only want the pathway names.

```
# TODO
```

Question 3

Use Hallmark GSEA results to compare positive and negative enrichment.

Tasks:

- create `hallmark_positive` using `NES > 0`
- create `hallmark_negative` using `NES < 0`
- print the first 5 descriptions from each table

Hint: Use `filter()` with conditions on `NES` to split positive and negative enrichment. Then inspect the `Description` column with `head()`.

```
# TODO
```

Short Answer

In 2 to 3 sentences, explain:

- what a positive NES usually suggests
- why GSEA results are often sorted by adjusted p value